COMPLETION ALONG A CLOSED SUBSPACE IS COMPATIBLE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The delivered context span is the author's own definition of the completion of an algebraic space along a closed subspace; it is copied whole, including the author's own reference to the colimit lemma that makes the object a formal algebraic space.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Completion of an algebraic space along a closed subspace, as the colimit of the infinitesimal neighbourhoods. Same chapter, Definition definition-completion-subspace, invoked at proof line 8255. Supplies the objects (X')^_{Z'} and X^_Z and the colimit formulas used at the end of the proof.
- Morphisms of algebraic spaces representable by algebraic spaces, tested on schemes. Bootstrap, the sections on morphisms representable by spaces and their properties. Lets the cartesian property be tested on morphisms Y -> X from a scheme and gives the meaning of the final 'in particular' clause.

Source reference. Original lemma 0GVN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7903–7903. Pointer outside the counted span, invoked only from a delivered context span as a forward reference; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let X be an algebraic space over S. Let $Z \subset X$ be a closed subspace and denote $Z_n \subset X$ the nth order infinitesimal neighbourhood. The formal algebraic space

$$X^\wedge_Z = \operatorname{colim} Z_n$$

(see Lemma 0GVN) is called the *completion of X along Z*.

Lemma 0.2. *Let S be a scheme. Let $f : X' \to X$ be a morphism of algebraic spaces over S. Let $Z \subset X$ be a closed subspace and let $Z' = f^{-1}(Z) = X' \times_X Z$.*

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Then

$$\begin{array}{ccc} (X')^\wedge_{Z'} & \longrightarrow & X' \\ \downarrow & & \downarrow f \\ X^\wedge_Z & \longrightarrow & X \end{array}$$

is a cartesian diagram of sheaves. In particular, the morphism $(X')^\wedge_{Z'} \to X^\wedge_Z$ is representable by algebraic spaces.

Proof. Namely, suppose that $Y \to X$ is a morphism from a scheme into X such that $Y \to X$ factors through Z. Then $Y \times_X X' \to X$ is a morphism of algebraic spaces such that $Y \times_X X' \to X'$ factors through Z'. Since $Z'_n = X' \times_X Z_n$ for all $n \geq 1$ the same is true for the infinitesimal neighbourhoods. Hence the cartesian square of functors follows from the formulas $X^\wedge_Z = \operatorname{colim} Z_n$ and $(X')^\wedge_{Z'} = \operatorname{colim} Z'_n$. $\square$